Phase Transition in Nonparametric Minimax Rates for Covariate Shifts on Approximate Manifolds

Yuyao Wang[1], Nabarun Deb[2], Debarghya Mukherjee[1]

[1]Department of Mathematics and Statistics, Boston University

[2]Booth School of Business, University of Chicago

Outline

1. Introduction
2. Contributions
3. Problem Setup
4. Main Results
5. Adaptive Estimation
6. Numerical Experiments: Non-Adaptive Estimation
7. Numerical Experiments: Adaptive Estimation
8. Conclusion

Introduction

- In many real-world scenarios, target data is scarce, while abundant auxiliary (source) data is available.
- Transfer learning leverages relevant information across tasks to improve inference.
- We focus on *covariate shift*: covariate distributions differ, conditional distributions remain invariant.
- Motivated by manifold hypothesis: target lies on an approximate low-dimensional manifold inside source.
- Goal: establish minimax rates and identify phase transitions under geometric + smoothness constraints.

Main Contributions

- **Setup:** Source samples n_P, target samples n_Q, ambient dimension D, intrinsic dimension d, manifold deviation ρ_n, smoothness β.
- **Phase Transition:**
  - If $\rho_n \gg \kappa_n^*$, minimax rate depends on ρ_n:

$$\left(n_P + n_Q \rho_n^{d-D}\right)^{-\frac{2\beta}{2\beta+D}}.$$

  - If $\rho_n \ll \kappa_n^*$, rate matches ideal manifold case:

$$\left(n_P^{\frac{2\beta+d}{2\beta+D}} + n_Q\right)^{-\frac{2\beta}{2\beta+d}}.$$

- **Implications:**
  - Source data accelerates rates beyond target-only estimation.
  - Faster than naïve pooling in D dimensions.
- **Adaptive Estimator:**
  - Local polynomial regression with data-driven choice of β, d.
  - Nearly optimal rates (up to logs).

Problem Setup

- **Data:**
  - Source: $(X_i, Y_i) \sim P = P_X \times P_{Y|X}$, n_P samples
  - Target: $(X_i, Y_i) \sim Q = Q_X \times Q_{Y|X}$, n_Q samples
- **Covariate Shift Assumption:**

$$P_X \neq Q_X, \quad \mathbb{E}_P[Y|X] = \mathbb{E}_Q[Y|X] = f^\star(x).$$

- **Goal:** Estimate $f^\star(x^\star)$ at a target point $x^\star$.

Modeling Assumptions

- **Approximate Manifold:**
  - Target covariates lie on $\mathcal{M}_{\rho_n}$:

$$X = \phi(V) + \rho_n U, \quad V \in [-1,1]^d, \; U \in [-1,1]^D.$$

  - $\rho_n = 0$: exact manifold; $\rho_n > 0$: noisy embedding.
- **Regularity:**
  - Source density $p(x)$ positive, Lipschitz near $x^\star$.
  - ϕ bi-Lipschitz $\implies$ smooth manifold geometry.
- **Smoothness of $f^\star$:**

$$f^\star \in \Sigma(\beta, L) \quad \text{(locally } \beta\text{-Hölder)}.$$

Estimator: Local Polynomial Regression

- Fit a polynomial of degree $\ell = \lfloor \beta \rfloor$ locally around $x^\star$.
- Kernel-weighted least squares:

$$\hat{f}^{\mathrm{LPR}}(x^\star) = e_1^\top (Z^\top W Z)^{-1} (Z^\top W Y) = \sum_{i=1}^{n} w_i Y_i.$$

- Properties:
  - Weights reproduce low-order polynomials.
  - Less bias near boundary than kernel regression.
- **Truncation for stability:** Use $\hat{f}(x^\star) = 0$ if $\lambda_{\min}(Z^\top W Z)$ is too small.

Upper Bound: Phase Transition

Theorem (Upper Bound). For LPR estimator under covariate shift, we have

$$\mathbb{E}(\hat{f}(x^\star) - f^\star(x^\star))^2 \leq \begin{cases} C\left(n_P + n_Q \rho_n^{d-D}\right)^{-\frac{2\beta}{2\beta+D}}, & \rho_n \gtrsim \kappa_n^*, \\ C\left(n_P^{\frac{2\beta+d}{2\beta+D}} + n_Q\right)^{-\frac{2\beta}{2\beta+d}}, & \rho_n \lesssim \kappa_n^*. \end{cases}$$

- Bandwidth h_n chosen adaptively.
- Shows a **phase transition** at $\rho_n \asymp \kappa_n^*$.
- Bias–variance analysis under manifold structure.

Insights from Upper Bound

New rates for target-only:

$$\mathbb{E}(\hat{f} - f^\star)^2 \lesssim \begin{cases} (n_Q \rho_n^{d-D})^{-\frac{2\beta}{2\beta+D}}, & \rho_n \gtrsim n_Q^{-\frac{1}{2\beta+d}}, \\ n_Q^{-\frac{2\beta}{2\beta+d}}, & \rho_n \lesssim n_Q^{-\frac{1}{2\beta+d}}. \end{cases}$$

- Reveals a **novel phase transition** at $\rho_n \asymp n_Q^{-1/(2\beta+d)}$.
- If ρ_n small: same rate as noiseless manifold case.
- If ρ_n large: degrade to D-dimensional nonparametric rate.

Benefit of Transfer:

- Effective sample size $n_{\text{eff}} = n_P^{(2\beta+d)/(2\beta+D)} + n_Q$.
- Rate faster than target-only $n_Q^{-2\beta/(2\beta+d)}$.
- Faster than naïve pooled $(n_P + n_Q)^{-2\beta/(2\beta+D)}$.

Lower Bound and Optimality

Theorem (Lower Bound). For any estimator $\hat{f}$,

$$\inf_{\hat{f}} \sup_{f \in \Sigma(\beta, L)} \mathbb{E}(\hat{f}(x^\star) - f(x^\star))^2 \geq \begin{cases} C\left(n_P + n_Q \rho_n^{d-D}\right)^{-\frac{2\beta}{2\beta+D}}, & \rho_n \gtrsim \kappa_n^*, \\ C\left(n_P^{\frac{2\beta+d}{2\beta+D}} + n_Q\right)^{-\frac{2\beta}{2\beta+d}}, & \rho_n \lesssim \kappa_n^*. \end{cases}$$

- Matches the upper bound $\Rightarrow$ **minimax optimal**.
- Same bandwidth h_n works across regimes.
- Learner does not need to distinguish source vs. target samples.

Adaptive Estimation: Motivation

- Our LPR estimator requires knowledge of:

$$d \quad \text{(intrinsic dimension)}, \quad \beta \quad \text{(smoothness index)}.$$

- In practice, both are unknown.
- Goal: construct a **data-driven adaptive estimator** $\hat{f}_{\text{adp}}$.
- Focus on regime $\rho_n \ll \kappa_n^*$ (target lies close to manifold).

Step 1: Intrinsic Dimension Estimation

- Use k-nearest neighbor method.
- Idea: in d-dim space, number of neighbors grows like r^d.
- Local estimate at X_i:

$$\hat{d}(X_i) = \frac{\log 2}{\log\left(r^{(k)}(X_i)/r^{(\lceil k/2 \rceil)}(X_i)\right)}.$$

- Aggregate across sample:
  - Average: $\hat{d}_{\text{avg}}$,
  - Majority vote: $\hat{d}_{\text{vote}}$.
- Both converge to true d exponentially fast.

Step 2: Adaptive Smoothness via Lepski's Method

- Bandwidth h depends on β; trade-off bias vs. variance.
- Construct candidate set $\widetilde{\mathcal{B}} = \{\tilde{\beta}_1, \ldots, \tilde{\beta}_N\}$.
- For each $\tilde{\beta}$, compute estimator $\hat{f}_{h_{n,\tilde{\beta}}}$.
- Lepski rule: choose largest $\tilde{\beta}$ such that estimates remain stable:

$$|\hat{f}_{h_{n,\tilde{\beta}}} - \hat{f}_{h_{n,\eta}}| \leq C_\ell h_{n,\eta}^\eta, \quad \forall \eta \leq \tilde{\beta}.$$

- Final estimator: $\hat{f}_{\text{adp}} = \hat{f}_{h_{n,\hat{\beta}}}$.

Guarantee for Adaptive Estimator

Theorem. If $\rho_n \ll \kappa_n^*$, then

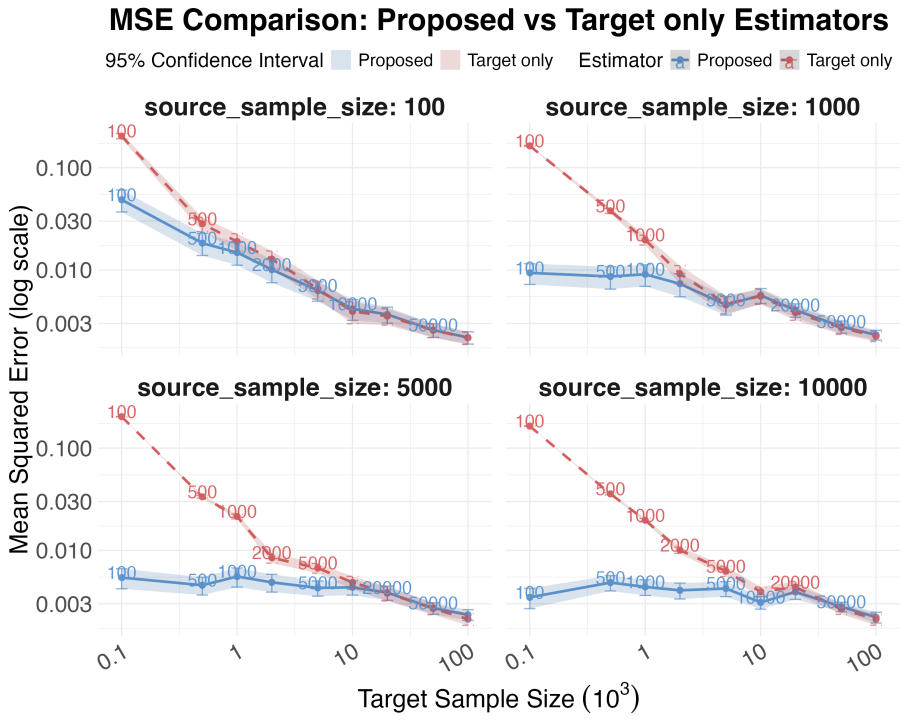
$$\mathbb{E}\left[|\hat{f}_{\text{adp}}(x^\star) - f^\star(x^\star)|^2\right] = O\left(\left(\frac{n_Q + n_P^{\frac{2\beta+d}{2\beta+D}}}{\log n}\right)^{-\frac{2\beta}{2\beta+d}}\right).$$

- Achieves minimax rate up to a log factor.
- Log penalty is unavoidable in smoothness adaptation.
- With $\hat{d}_{\text{avg}}$ or $\hat{d}_{\text{vote}}$, also adaptive to dimension.

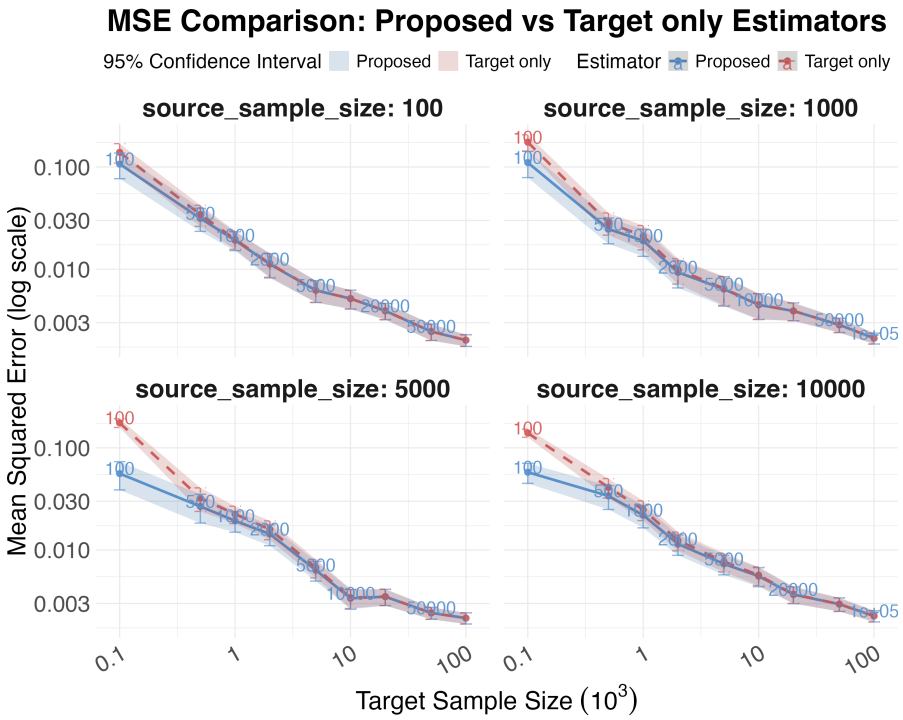
Experiment Setup: Same Dimension ($D = 5$)

- **Goal:** Compare *target-only* vs. *proposed* estimator.
- **Data generation:**
  - Source: $X \sim \mathrm{Unif}([-0.5, 0.5]^5)$
  - Target: $X \sim \mathrm{Unif}([0, 1]^5)$
  - $Y = f^\star(X) + \epsilon,\ \epsilon \sim \mathcal{N}(0, 1)$
  - $f^\star(x) = \sum_{i=1}^{5} |x_i - x_i^*|^{2.5}$
- **Evaluation points:**
  - Interior: x^*_{Int} (inside both supports)
  - Target-only: x^*_{to} (outside source support)

Results: Mean Squared Error (MSE)



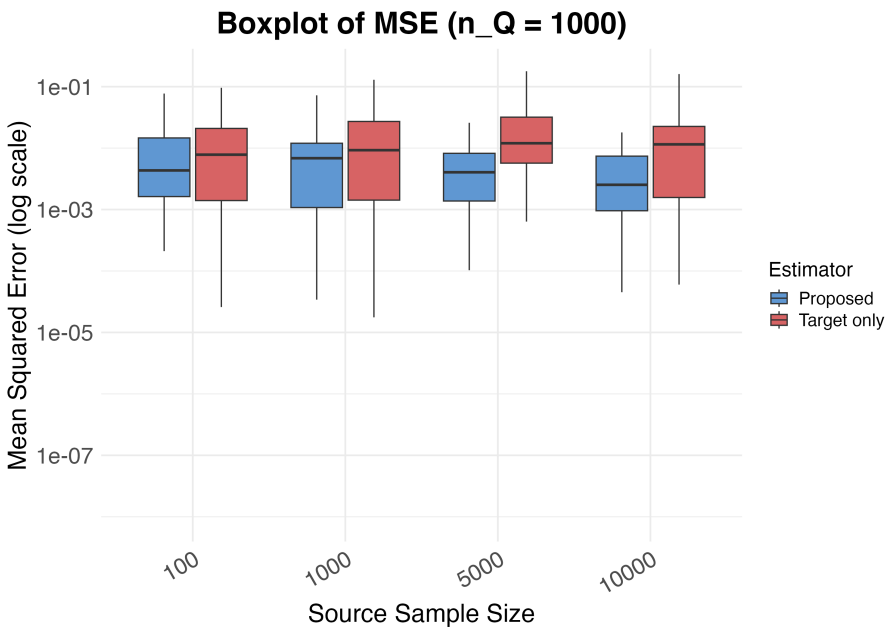
MSE at x^*_{Int}



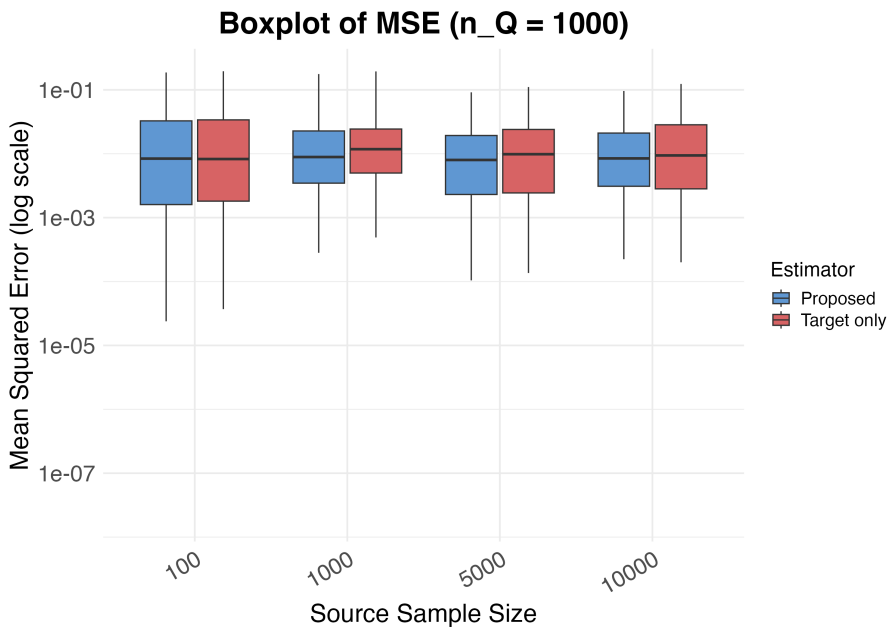
MSE at x^*_{to}

- Interior point: proposed estimator $\ll$ target-only, especially for small n_Q.
- Target-only point: both estimators coincide $\Rightarrow$ no negative transfer.

Results: Boxplots ($n_Q = 1000$)



Interior point



Target-only point

- Interior: lower median MSE and variance; improves as n_P increases.
- Target-only: matches target-only estimator; robust to negative transfer.

Experiment Setup: Manifold Case ($D = 5, d = 2$)

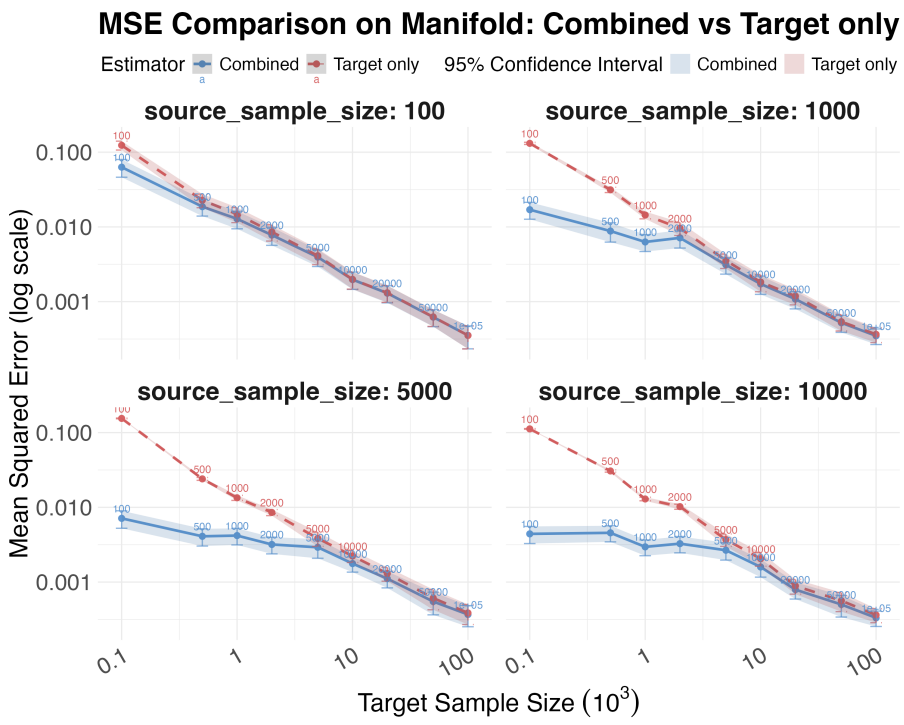
- **Goal:** Compare *target-only* vs. *proposed* estimator when target lies on a manifold.
- **Data generation:**
  - Source: $X \sim \mathrm{Unif}([0,1]^5)$
  - Target: $X = \phi(Z),\ Z \sim \mathrm{Unif}([-1,1]^2)$
  - Embedding:

$$\phi(z_1, z_2) = \left(\tfrac{z_1+1}{2},\ \tfrac{z_2+1}{2},\ z_1^2,\ z_2^2,\ \tfrac{(z_1+1)(z_2+1)}{4}\right)$$

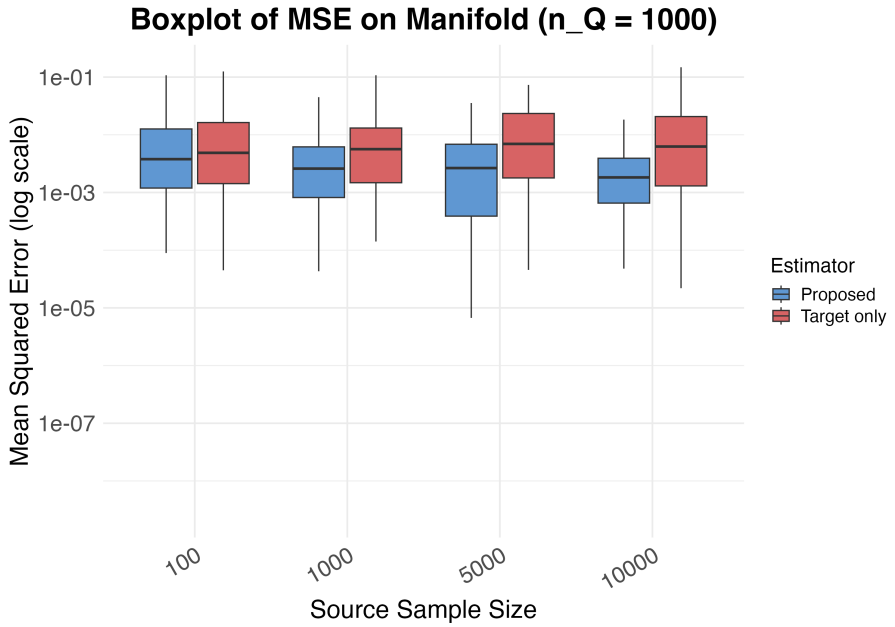
  - $Y = f^\star(X) + \epsilon,\ \epsilon \sim \mathcal{N}(0,1)$, with $f^\star$ β-Hölder smooth, $\beta = 2.5$
- **Evaluation point:**

$$x^* = \phi((-0.4248,\ 0.5766)) = (0.2876,\ 0.7883,\ 0.1805,\ 0.3325,\ 0.2267)$$

Results: MSE vs. Target Sample Size



MSE vs. n_Q



Boxplot ($n_Q = 1000$)

- Proposed estimator $\ll$ target-only when n_Q is small.
- Gap decreases as n_Q grows (source info less needed).
- Boxplot shows: lower median error and reduced variance.

Adaptive Estimation: Setup

- **Goal:** Estimate d (intrinsic dimension) and β (smoothness) from data.
- **Source data:** $X \sim \mathrm{Unif}([0,1]^5)$
- **Target data:** $(Z_1, Z_2) \sim \mathrm{Unif}([-1,1]^2)$, embedded via

$$\phi(z_1, z_2) = \left(\tfrac{z_1+1}{2},\ \tfrac{z_2+1}{2},\ z_1^2,\ z_2^2,\ \tfrac{(z_1+1)(z_2+1)}{4} \right).$$

- **Regression:** $Y = f^\star(X) + \epsilon$, $\epsilon \sim \mathcal{N}(0,1)$
- **Test point:** $x^* = \phi(-0.4248,\ 0.5766)$

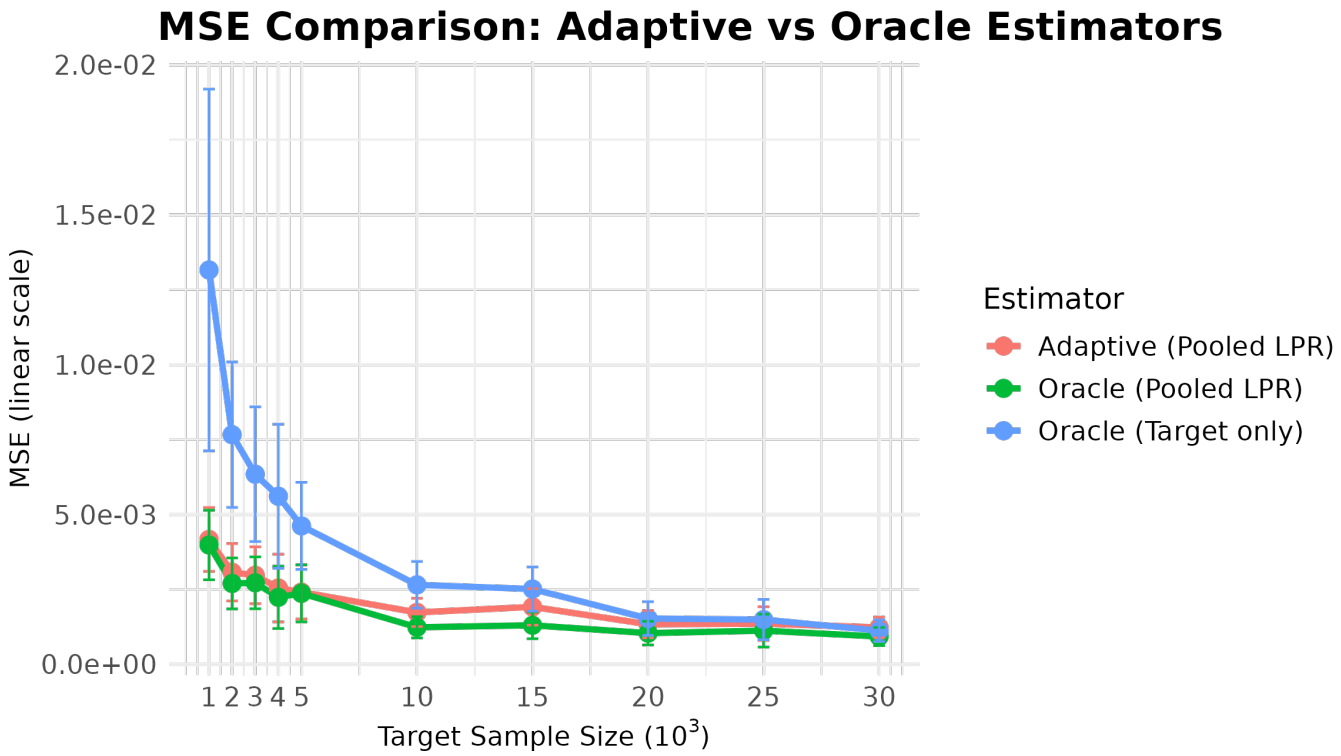
Estimators Compared

- **Adaptive (Pooled LPR)**:
  - Estimate d via k-NN ($k = 10$)
  - Select β via Lepski's method
- **Oracle (Pooled LPR)**: true (d, β), uses source + target data
- **Oracle (Target only)**: true (d, β), only target data

Experiment setup:

- Fix $n_P = 5000$
- Vary $n_Q \in \{1000, 2000, \ldots, 30000\}$
- 100 Monte Carlo repetitions

Results: Adaptive Estimation



- Adaptive estimator $\approx$ Oracle (Pooled LPR): robust and accurate
- Large gain over Oracle (Target only) when n_Q is small
- Confirms benefit of transfer learning under covariate shift

Key Contributions

- **Theory:** Established minimax rates under covariate shift with approximate manifolds
  - Identified a **phase transition** depending on noise level ρ_n
  - Rates adapt to both ambient dimension D and intrinsic dimension d
- **Estimator:** Local Polynomial Regression (LPR)
  - Achieves minimax optimal rates
  - Robust to covariate shift and manifold noise
- **Adaptive Method:**
  - d estimated via k-NN
  - β selected via Lepski's method
  - Nearly optimal up to a log factor

Takeaways and Future Work

- **Numerical experiments:**
  - Proposed estimator consistently outperforms target-only baseline
  - No negative transfer observed when supports do not overlap
  - Adaptive estimator $\approx$ Oracle in practice
- **Broader impact:**
  - Provides a principled framework for transfer learning in structured, high-dimensional settings
  - Opens the door to applications in vision, NLP, medical diagnosis
- **Future directions:**
  - Extension to classification and sequential decision-making
  - Deeper exploration of manifold adaptivity with multiple sources

References

Fan, J. (1992).
Design-adaptive nonparametric regression.
Journal of the American Statistical Association, 87(420), 998–1004.

Fan, J., Heckman, N. E., & Wand, M. P. (1995).
Local polynomial kernel regression for generalized linear models and quasi-likelihood functions.
Journal of the American Statistical Association, 90(429), 141–150.

Farahmand, A. M., Szepesvári, C., & Audibert, J.-Y. (2007).
Manifold-adaptive dimension estimation.
In *Proceedings of the 24th International Conference on Machine Learning (ICML '07)*, 265–272. ACM.

Lepski, O. V., & Spokoiny, V. G. (1997).
Optimal pointwise adaptive methods in nonparametric estimation.
The Annals of Statistics, 25(6), 2512–2546.

Lepskii, O. (1991).
On a problem of adaptive estimation in Gaussian white noise.
Theory of Probability & Its Applications, 35(3), 454–466.